Emran Yasser Moustafa

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PhD researcher at the University of Edinburgh with a first-class engineering background across robotics, AI, embedded hardware and autonomous vehicles. Experienced in assessing emerging technologies for industry as a consultant, and in founding and running organisations. Keen interest in product development and commercialisation.

Education

2025 – pres. **Ph.D. in Informatics** — University of Edinburgh

Thesis: *"Shared Autonomy in Assistive Driving through Online Action-Space Shaping"*.

Developing advanced driver assistance tools, including those for shared autonomy in interaction-rich scenes. I am particularly interested in planning and control for human-robot interaction. Currently, I am working on a new framework for shared control that better supports assistance in settings with high uncertainty about the operator's intent.

Supervisors: Prof. Subramanian Ramamoorthy (Edinburgh), Prof. Gustav Markkula (Leeds).

2020 – 2025 **B.Eng. & M.Eng. Electronic & Computer Eng.** — Trinity College Dublin

- First Class Honours; Gold Medal for academic excellence.
- M.Eng. thesis on model-based reinforcement learning for autonomous racing, supervised by Prof. Ivana Dusparic. Awarded the **Collen Prize** for the best Computer Engineering thesis; published at **IEEE ICAR 2025**.
- Exchange year at **TU Delft** (M.Sc. Robotics modules in swarm robotics and 3D perception). B.Eng. thesis on trajectory prediction in mixed traffic with Prof. Julian Kooij, published in **IEEE RA-L**.

Experience

2026 **Consultant** — Copenhagen Group A/S

Advised on AI for intelligence, surveillance and reconnaissance for the Danish Navy. Assessed the technical feasibility and maturity of emerging technologies and reported recommendations for commercial stakeholders. Contracted April – September 2026.

2022 – 2025 **Research Internships** — TCD · TU Delft · Flying Labs

- **Trinity College Dublin (2025):** model-based reinforcement learning for autonomous racing and traffic signal control.
- **TU Delft, Lab Assistant (2024):** developed a ROS framework for drone swarm control; supported public engagement projects communicating robotics research.
- **Flying Labs Madrid (2023):** built a visual SLAM pipeline for micro-air vehicles.
- **Trinity College Dublin (2023):** developed a ROS framework for drone swarm control; used for coursework at Trinity College Dublin (Ireland), Lund University (Sweden) and KAIST (Korea).
- **Trinity College Dublin (2022):** hardware-accelerated perception for autonomous racing.

Leadership & Enterprise

2025 – pres. **Co-founder** — Do Sheal

Co-founded an indie board game studio in Dublin developing real-world-inspired strategy games, covering product development through to bringing designs to market. Received EU funding.

2022 – 2023 **Founder** — RacingAI

Founded a non-profit to grow Ireland's robotics and autonomous navigation community, running events across Ireland for 18 months, including intervarsity robotics competitions. Received EU funding.

2021 – 2023 **Driverless Team Captain** — Formula Trinity

Led the autonomous racing team through **F1Tenth 2021** and **Formula Student UK 2022–23**. Nominated by teammates for the Craig Dawson Award (MVP) in 2022. Multiple team awards during my tenure.

Publications

First author

- *Context-Aware Model-Based Reinforcement Learning for Autonomous Racing.* **EYM**, Ivana Dusparic. **ICAR 2025**.

Co-author

- *Road User Specific Trajectory Prediction in Mixed Traffic Using Map Data.* Hidde Boekema, **EYM**, Julian Kooij, Dariu Gavrila. **RA-L 2025**.

Honours

2025	**UKRI PhD Scholarship**	University of Edinburgh
2025	**Collen Prize for Computer Engineering (best M.Eng. thesis)**	Trinity College Dublin
2025	**Gold Medal, Electronic & Computer Engineering**	Trinity College Dublin
2023	**Stanford-Smith Prize (highest overall B.Eng. grades)**	Trinity College Dublin
2023	**Dean of Students' Leadership Award for Volunteering**	Trinity College Dublin
2022	**Laidlaw Leadership & Research Scholarship (€10,000)**	Laidlaw Foundation
2021	**Eric & Barbara Kinsella Scholarship (€20,000)**	Jones Engineering
2021	**E.R. Stuart Prize for Engineering**	Trinity College Dublin

Skills

Technical ML & reinforcement learning, robotics, MPC, game theory.

Tools Python, C/C++, ROS, PyTorch, Gym, Autoware, CARLA.

Training pro^2 Device Prototyping Summer School, Lancaster University (2026).

Leadership Team captaincy, project management, event organisation, founding and running organisations.